

DynaDrive-V0.1

Smart Closed-Loop DC Motor Controller.

Description

DynaDrive is a sophisticated, autonomous DC motor controller system engineered for enhanced operational safety and reliability. Its core functionality revolves around a closed-loop control scheme that intelligently modulates motor power by utilizing real-time data from an integrated digital temperature and humidity sensor. This preemptive thermal management system is designed to prevent motor degradation and catastrophic failure. The architecture is supported by a robust dual power supply, isolating the motor's high-voltage circuits from the sensitive control electronics to ensure stable, noise-free operation of the main MCU, which is easily programmable via an onboard USB-to-UART interface.

Designed for modern, interconnected applications, DynaDrive's capabilities can be significantly extended through its dedicated external pins for an ESP32 Wi-Fi module. This expansion enables comprehensive remote management, allowing users to control motor power and monitor real-time temperature and humidity data over the internet with flexibility. For immediate local diagnostics, the system provides clear visual feedback through dedicated LEDs that indicate power supply status and a dynamic RGB LED that varies its color in correspondence with the motor's operating temperature, offering an at-a-glance assessment of the system's condition.

Resources

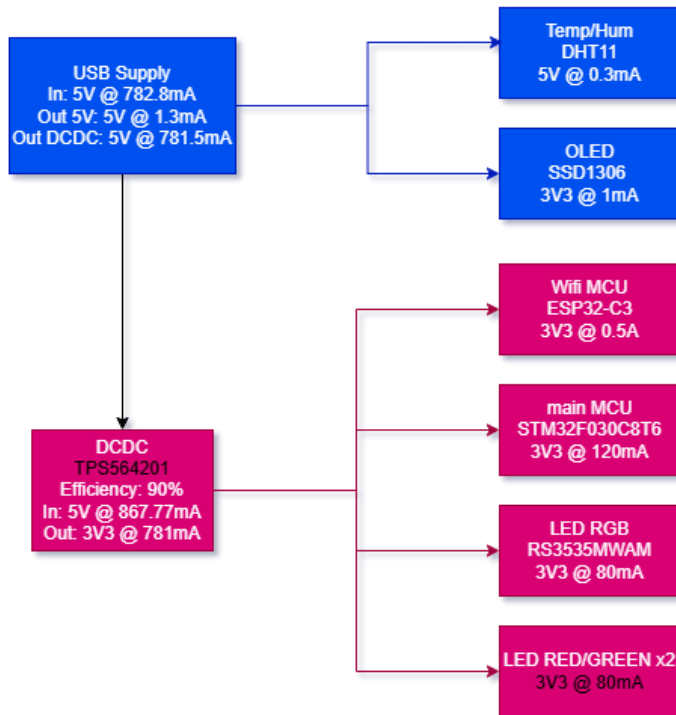
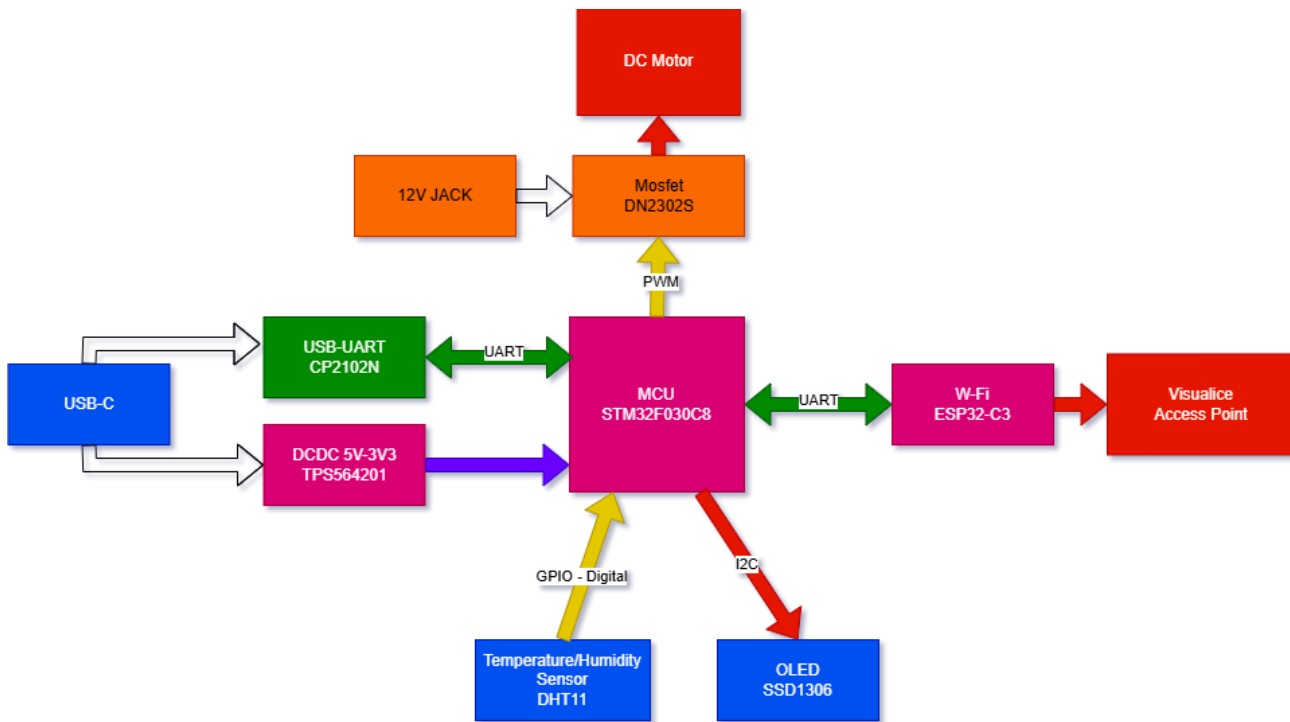
CP2102N	USB to UART controller
DHT11	Temp/Hum Sensor
STM32F030C8T6	main MCU
ESP32-C3-WROOM02	MCU(Wi-Fi)
SSD1306	128x64 OLED
TPS564201	DCD 5V->3V3
XL-2012UGC	LED GREEN
RS-3535MWAM	RGB LED
DN2302S	Mosfet N-Channel

- USB 2.0 (compatible speed) with integrated USB-to-UART driver. Also 5V 1A USB supply power
- Temperature and humidity digital sensor, low-cost, fast response, 16bit resolution.
- ESP32 with Wi-Fi (802.11b/g/n up to 150 Mbps) and Bluetooth v5.
- CMOS OLED/PLED driver with controller for organic / polymer light emitting diode dot-matrix graphic display system 182 x 64 pixels.
- Arm® 32-bit Cortex®-M0 CPU, frequency up to 48 MHz, 4 to 32 MHz crystal oscillator
- 12V, 2A external power supply connector.
- Dual full-bridge driver with up to 46V operating supply, 4A DC current, low saturation voltage, overtemperature protection, and high noise immunity
- Functional safety-capable DCDC converter with 2.7-6V input, 0.6-5.5V output, up to 2A continuous current, and 1% accuracy.
- Various LEDS for better user info system

Applications

- Motor Control System
- Industrial Automation
- Thermal Management
- Remote Monitoring & Control
- Predictive Maintenance

Features



References

- [1] J. Cook, "Simple DC motor speed control with PWM," Arrow Electronics, Inc., Research and Events Articles, 23 Mar. 2023. [Online]. Available: <https://www.arrow.com/en/research-and-events/articles/simple-dc-motor-speed-control-with-pwm> [Accessed: 12-Jun-2025]. (arrow.com)
- [2] U. Subba, "Speed control of DC motor with PWM using Arduino," ee-diary, 15-Sep-2020. [Online]. Available: <https://www.ee-diary.com/2020/09/Speed-control-DC-motor-PWM-Arduino.html> [Accessed: 12-Jun-2025]. (ee-diary.com)